

Project 01 Requirements

Student Score Analyzer



Project Description

Create a JavaScript program that analyzes a list of student scores. The program should determine the grade for each score and calculate the average score, the number of passed students, and the number of failed students.



Input

Use the following list of student scores:

```
const scores = [85, 42, 73, 96, 58];
```



Grading Rules

Determine the grade for each score using the following rules:

Score Range	Result
90 - 100	Excellent
70 - 89	Good
50 - 69	Pass
Below 50	Fail



Requirements

The program should:

- 1 Check each score in the list.
- 2 Determine the grade based on the rules.
- 3 Calculate the average score.
- 4 Count the number of passed students (scores 50 or higher).
- 5 Count the number of failed students (scores below 50).
- 6 Display the grade for each score and the final statistics.



Expected Output

The program should display the grade for each score and then show the final statistics. The output should look like this:

```
Score: 85 - Good
Score: 42 - Fail
Score: 73 - Good
Score: 96 - Excellent
Score: 58 - Pass
-----
Average: 70.8
Passed: 4
Failed: 1
```

Project 01 Code & Output

Student Score Analyzer



```
JS student_Score_Analyzer.js x
1  const scores = [85, 42, 73, 96, 58];
2
3  function getGrade(score) {
4      if (score >= 90) {
5          return "Excellent";
6      } else if (score >= 70) {
7          return "Good";
8      } else if (score >= 50) {
9          return "Pass";
10     } else {
11         return "Fail";
12     }
13 }
14
15 let total = 0;
16 let passedStudents = 0;
17 let failedStudents = 0;
18
19 for (let i = 0; i < scores.length; i++) {
20     const score = scores[i];
21     const grade = getGrade(score);
22     console.log(`Score: ${score} - ${grade}`);
23     total += score;
24     if (score >= 50) {
25         passedStudents++;
26     } else {
27         failedStudents++;
28     }
29 }
30
31 const average = total / scores.length;
32 console.log("-----");
33 console.log(`Average: ${average}`);
34 console.log(`Passed: ${passedStudents}`);
35 console.log(`Failed: ${failedStudents}`);
```

```
Terminal
PS D:\05_Mini_Projects> node
student_Score_Analyzer.js

Score: 85 - Good
Score: 42 - Fail
Score: 73 - Good
Score: 96 - Excellent
Score: 58 - Pass
-----
Average: 70.8
Passed: 4
Failed: 1
```

